

Supplementary material (Online Resource 1): the derivations to the article “Universe24: one integer rule on a cubic lattice, read through clicks, gives the wave equation, the quantum of action and a weak-field metric as three readings of one line”, submitted to Foundations of Physics

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October 2026

This supplement gives the algebraic steps of every result the paper marks DERIVED or COMPUTED without a proof in its text, in the order of the paper’s sections; the theorems proved in the paper are not repeated. Each derivation names its inputs from the paper, its steps, and, where a number was computed, the computation and its check. The notation is the paper’s: Rule3’s coefficients at a Node of clock pace p_0 and Link paces p_a are $R_a = 2 \text{ num } p_a^2$ on the two arrivals of the axis a , $S = 12 \text{ den } \Gamma^2 - 12(\text{den} - \text{num}) p_0^2 - 4 \text{ num } \sum_a p_a^2$ on the Node’s own level, and the wall $w = 6 \text{ den } \Gamma^2$; the content read at the Node is $x = c/\Gamma$, also written U ; $g = (\text{den} - \text{num})/\text{den} = 1 - \cos \omega_0$.

The path from Rule3 to each result

Every result of the paper is reached from Eq. (1), Rule3’s line with its one write and its three readings, by a chain of the derivations below; where a link of a chain is an assumption of the model, it is named in the chain, so that each path is closed in the algebra with its assumptions explicit.

- *Light’s speed and dispersion*: Eq. (1) at the vacuum’s pace $\rightarrow$ the band, S.1.
- *Gravity as a slowed clock, light’s index, the bending and Shapiro’s delay*: Eq. (1) $\rightarrow$ the paces, Eq. (3) (the assumption: the level read once into the clock and twice into the Link) $\rightarrow$ the band at a pace, S.1 $\rightarrow$ the index, S.2 $\rightarrow$ the ray integral, Eq. (15) and S.20.
- *The redshift of a bound rotation*: Eq. (1) $\rightarrow$ S.1 at $k = 0$ with the clock’s pace $\rightarrow$ S.15.
- *The fall, the Newtonian potential the orbits measure and the post-Newtonian parameters*: Eq. (1) $\rightarrow$ S.1 at small k $\rightarrow$ the inertia and the ray equations, S.18 $\rightarrow$ S.19.
- *Lorentz’s factor to the second order*: Eq. (1) $\rightarrow$ the exact band of a free record, S.1 $\rightarrow$ the rotation at the moving centre, S.21, closed in the algebra for a free quantum of any family; for a body, the body’s own band from its fixed point, S.13, computed by the generator, enters the same S.21.
- *De Broglie’s relation*: Eq. (1) $\rightarrow$ S.1 $\rightarrow$ S.22.
- *From the Node to us, bodies, clusters and the observer*: Eq. (1) with the write $\rightarrow$ the fixed point and the functional, S.13 $\rightarrow$ the stability, Theorem 4 $\rightarrow$ the cloud and the free fall (computed) $\rightarrow$ the clicks between bodies, S.9, S.14 to S.17 $\rightarrow$ the clusters and the observer as a cluster (Section 7.5, with the assumption that the universe is bound).
- *The count*: Eq. (1) $\rightarrow$ the conserved form, Theorem 3 $\rightarrow$ the share, S.5 and S.6 $\rightarrow$ the invariant and the drift, S.14.
- *Light born by the write, absorption and emission*: Eq. (1)’s write line $\rightarrow$ the Wronskian, S.9 $\rightarrow$ the resonance, S.16.
- *Bell*: Eq. (1) linear $\rightarrow$ the parts’ determinism (Section 9.4) $\rightarrow$ the declared credit (the assumption) $\rightarrow$ S.10.
- *The two slits*: Eq. (1) stepped in real arithmetic on the declared world $\rightarrow$ S.23 and the computed numbers of Section 9.
- *The constants and the bounds*: Eq. (1) $\rightarrow$ S.1 $\rightarrow$ S.24 (the Link); the field’s rest $\rightarrow$ S.25 (Newton’s constant); S.11 (the reach), S.12 (the two speeds).

The model (Section 3 of the paper)

Derivation S.1 (the band at a pace, Eq. (4) of the paper). Inputs: *Rule3’s line at the vacuum’s paces, where the two Links of an axis share one $R_a = 2 \text{ num } \Gamma^2$, $w a_{\text{next}} = \sum_a R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}}$, with the remainders dropped (they are bounded by w and do not move a plane wave’s rotation).* Steps: *for a plane wave*

$a_t(\mathbf{x}) = A \cos(\mathbf{k} \cdot \mathbf{x} - \omega t)$ the two arrivals of the axis a sum to $2 \cos k_a a_{\text{now}}$ and $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$, so

$$2 \cos \omega = \frac{2 \sum_a R_a \cos k_a + S}{w} = \frac{4 \text{num} \sum_a p_a^2 \cos k_a + 12 \text{den} \Gamma^2 - 12(\text{den} - \text{num}) p_0^2 - 4 \text{num} \sum_a p_a^2}{6 \text{den} \Gamma^2},$$

that is $\cos \omega = 1 - g(p_0/\Gamma)^2 - \frac{\text{num}}{3 \text{den}} \sum_a (p_a/\Gamma)^2 (1 - \cos k_a)$, which is Eq. (4). At the vacuum's pace, $p_0 = p_a = \Gamma$, this is $\cos \omega = \text{num}/\text{den} - \frac{\text{num}}{3 \text{den}} \sum_a (1 - \cos k_a) = \frac{\text{num}}{3 \text{den}} \sum_a \cos k_a$; at $k = 0$, $\cos \omega_0 = \text{num}/\text{den}$. Light's speed and dispersion: for $[1, 1]$ along one axis $\cos \omega = (2 + \cos k)/3$; expanding, $1 - \omega^2/2 + \omega^4/24 = 1 - k^2/6 + k^4/72$, so $\omega = (k/\sqrt{3})(1 - k^2/36 + \dots)$ and the group velocity $d\omega/dk = \sin k/(3 \sin \omega) = (1/\sqrt{3})(1 - k^2/12 + \dots)$: the speed $1/\sqrt{3}$ at long wavelength and the group speed falling as $1 - 0.083 k^2$. Check: at $k = 0.05, 0.1$ and 0.2 the exact $\sqrt{3} d\omega/dk - 1$ over k^2 is $-0.0833, -0.0834$ and -0.0835 .

Derivation S.2 (light's index at a content, Section 3.4). Inputs: Eq. (4) with the paces of Eq. (3), $p_0 = \Gamma(1 - 1/\Gamma)^L = \Gamma N$, $N = e^{-x} + O(1/\Gamma)$ at the content $x = L/\Gamma$, and $p_a = p_0^2/\Gamma = \Gamma N^2$ for a reader at the weight 1 and no tension. Steps: for light, $g = 0$, so $\cos \omega = 1 - \frac{1}{3} N^4 \sum_a (1 - \cos k_a)$, and at long wavelength $\omega = N^2 |\mathbf{k}|/\sqrt{3}$. Light's speed at the content x is $N^2 = e^{-2x}$ times its vacuum speed, so its index is $n = e^{2x} = 1/(1 - 2x) + O(x^2)$ and $n - 1 = 2x + O(x^2)$: the Link's pace is the clock twice, and so is light's slowing.

Derivation S.3 (the guard's edge and the horizon, Section 3.4). Inputs: Eq. (4). Steps: the lowest rotation of the band is at $k_a = \pi$ on every axis, where $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2 - \frac{4 \text{num}}{3 \text{den}} \sum_a (p_a/\Gamma)^2$; the recurrence $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$ has no exponentially growing solution iff $|2 \cos \omega| \leq 2$; at the edges ± 2 a linearly growing solution exists ($u_t = \pm t$ from $u_0 = 0, u_1 = 1$), which the guard admits, light's checkerboard mode sitting at -2 ; for a negative numerator the lowest rotation is at $k = 0$ and the same bound holds with $|\text{num}|$. With $p_0 = p_a = p$ the condition $2 \cos \omega \geq -2$ reads $(p/\Gamma)^2 (2g + 4 \text{num}/\text{den}) \leq 4$, that is $(p/\Gamma)^2 \leq 2 \text{den}/(\text{den} + \text{num})$, so the edge is $P^2 = 2 \text{den} \Gamma^2 \text{div}(\text{den} + \text{num})$, which is Γ^2 for light. No horizon: under the composed paces $p_0 = \Gamma(1 - 1/\Gamma)^c > 0$ at every content and $p_a = p_0^2/\Gamma > 0$, so $R_a = 2 \text{num} p_a^2$ never vanishes and no Link closes; the quadratic truncation $p_a = \Gamma(1 - 2x)$ vanished at $x = 1/2$, a horizon of the truncation. The frozen clock: at a content c the clock runs at $N = (1 - 1/\Gamma)^c$ of the vacuum's, $e^{-c/\Gamma}$ to the unit, so a deep well freezes a clock to the unit and its Node's share is not a number there.

Derivation S.4 (the cube's group of 48 and its action, Section 3.1). Inputs: Rule3's line, whose six arrivals enter through the six Port coefficients R_{ij} , the two of an axis one $R_a = 2 \text{num} p_a^2$ with no coefficient naming an axis beyond its own pace; the readings D, F and W of Eq. (1). The group: a map of the six Ports onto themselves that keeps opposite Ports opposite is a signed permutation of the three axes, $\mathbf{x} \mapsto \mathbf{P}\mathbf{x}$ with $\mathbf{P}$ a permutation matrix with signs; there are $3! \cdot 2^3 = 48$ of them, the group O_h of the cube, of which the 24 with $\det \mathbf{P} = +1$ are the rotations and the other 24 reflections and rotatory reflections; with the translations of the lattice they form the space group of the simple cubic lattice. Covariance: let σ be such a map and let the turned world carry the levels $a_{\sigma(i)}^\sigma = a_i$ and the paces $p_{\sigma(a)}^\sigma(\sigma(i)) = p_a(i)$. The arrivals of the axis $\sigma(a)$ at $\sigma(i)$ in the turned world are the arrivals of the axis a at i in the original, in either order, and R_a is even in the axis's sign; so the sum $\sum_a R_a(\text{arr}_{+a} + \text{arr}_{-a})$ is the same number at $\sigma(i)$ as at i , S and w are the same, and the division gives the same quotient and remainder: the step commutes with σ , and a world turned by σ runs as the turned world. A translation commutes for the same reason. The readings' kinds: D_i is one number at a Node with no axis in it, carried to $\sigma(i)$ unchanged, a scalar. The currents F_{ij} are attached to the Links and carried with them, and the momentum density $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$ changes sign where σ reverses the axis a and is carried to the image axis otherwise, a vector; W_i carries no axis and is carried unchanged, and it changes sign under the reflection of a plane record's sense, $\text{im} \mapsto -\text{im}$, a pseudoscalar in that sense. The click (R, t) names a region and an interval and is carried to the turned region, a scalar. The time reversal: exchanging now and before leaves $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$ unchanged once next and before exchange their roles, since D is symmetric in the pair (next, before); it reverses F_{ij} and W_i , which are antisymmetric in (now, before); and it carries the step to its inverse by Theorem 1. The rational rotation: the two parts of a plane record obey the same linear line with the same coefficients, so any linear combination of them does; the combination $(\text{re}', \text{im}') = ((p\text{re} - q\text{im})/r, (q\text{re} + p\text{im})/r)$ with $p^2 + q^2 = r^2$ carries solutions to solutions, and its Wronskian is $(p^2 + q^2)/r^2$ times the old one, unchanged. What is not in the group: a boost changes the record's wave number and rotation together, which no permutation of Ports does, and a rotation by an angle other than the cube's carries a Node off the lattice; so neither is a symmetry, and Lorentz's factor, where the paper finds it, is a property of the clicks (Derivation S.21).

Exact properties (Section 4)

Derivation S.5 (the share's change is the currents, Eq. (10)). Inputs: the share at the vacuum's paces, $e_i = 3 \text{den}(\text{now}_i^2 + \text{before}_i^2) - \text{num} \text{now}_i S_6(\text{before})_i$, with S_6 the six-neighbour sum; Rule3 at the vacuum's paces, where $S = 0$, $R_a = 2 \text{num} \Gamma^2$ and $w = 6 \text{den} \Gamma^2$, so that $3 \text{den}(\text{next}_i + \text{before}_i) = \text{num} S_6(\text{now})_i + \rho_i$ with $\rho_i = (r_i - r'_i)/(2\Gamma^2)$ the

remainders' term (from $6 \text{ den } \Gamma^2(\text{next} + \text{before}) = 2 \text{ num } \Gamma^2 S_6(\text{now}) + r - r'$). Steps: after the step the pair (now, before) becomes (next, now), so

$$e'_i - e_i = 3 \text{ den } (\text{next}_i^2 - \text{before}_i^2) - \text{num} [\text{next}_i S_6(\text{now})_i - \text{now}_i S_6(\text{before})_i].$$

With $3 \text{ den } (\text{next}_i + \text{before}_i) = \text{num } S_6(\text{now})_i + \rho_i$ the first term is $(\text{next}_i - \text{before}_i)(\text{num } S_6(\text{now})_i + \rho_i)$, and the bracket cancels its next_i part:

$$\begin{aligned} e'_i - e_i &= \text{num} [\text{now}_i S_6(\text{before})_i - \text{before}_i S_6(\text{now})_i] + (\text{next}_i - \text{before}_i) \rho_i \\ &= \sum_{j \sim i} \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j) + (\text{next}_i - \text{before}_i) \rho_i, \end{aligned}$$

the six currents F_{ij} into the Node plus the remainder term. At any paces: with the weighted share $e_i = [w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i - \text{now}_i \sum_j R_{ij} \text{before}_j] / (2p_i^2)$ and the line without its division, $w \text{next}_i^* = \sum_j R_{ij} \text{now}_j + S_i \text{now}_i - w \text{before}_i$, the difference $e_i(\text{next}_i^*, \text{now}) - e_i(\text{now}, \text{before})$ is $\sum_j (q_{ij}^2 / \Gamma^2) \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ with no anisotropy term, the S_i terms and the Node's terms cancelling as in Theorem 3's proof, since $R_{ij} / (2p_i^2) = \text{num } q_{ij}^2 / \Gamma^2$ is the same from both ends. The integer step's $\text{next} = \text{next}_i^* + (r - r') / w$ adds $-(\text{next}_i - \text{before}_i)(r'_i - r_i) / (2p_i^2)$, because $w(\text{next}_i + \text{next}_i^*) - S_i \text{now}_i - \sum_j R_{ij} \text{now}_j = w(\text{next}_i - \text{before}_i)$. Check: exact fractions on a periodic chain of five Nodes at $[2, 3]$, $\Gamma = 10$, with contents 0 to 3, Link factors 5 to 10 and random levels: the identity holds at every Node, with and without the division; and the automated reviewer's witness, a uniform $\text{now} = 3$ over $\text{before} = 0$ at $\Gamma = 10$, $q = 5$, reads 81 per Node where the mean-of-paces form read 324 (an independent automated check).

Derivation S.6 (the total share is non-negative, Section 4.4). Inputs: e_i as above on a closed or periodic GameBoard; S_6 symmetric with eigenvalues $2 \sum_a \cos k_a \in [-6, 6]$. Steps: $\sum_i e_i = 3 \text{ den } (|\mathbf{n}|^2 + |\mathbf{b}|^2) - \text{num } \mathbf{n} \cdot S_6 \mathbf{b}$, and $|\mathbf{n} \cdot S_6 \mathbf{b}| \leq \|S_6\| |\mathbf{n}| |\mathbf{b}| \leq 6 |\mathbf{n}| |\mathbf{b}| \leq 3(|\mathbf{n}|^2 + |\mathbf{b}|^2)$, so $\sum_i e_i \geq 3(\text{den} - |\text{num}|)(|\mathbf{n}|^2 + |\mathbf{b}|^2) \geq 0$ for $|\text{num}| \leq \text{den}$. Equality needs $|\text{num}| = \text{den}$ and $\mathbf{b}$ an eigenvector of S_6 at ± 6 with $\mathbf{n} = \pm \mathbf{b}$: the static uniform level of light, or on an even periodic board its checkerboard, $\text{now} = (-1)^{x+y+z}$ with $\text{before} = -\text{now}$, whose shares are 0. A Node's share: at fixed before_i and $S_6(\text{before})_i = s$, e_i is a quadratic in now_i with the minimum $3 \text{ den } \text{before}_i^2 - \text{num}^2 s^2 / (12 \text{ den})$, negative iff $|s| > 6(\text{den}/\text{num}) |\text{before}_i|$; a plane record has $|s|/|\text{before}_i| = 2 \sum_a \cos k_a \leq 6$ and is never negative, a lay's edge can be. A region: pairing each boundary Link's cross term and bounding $2|\text{now}_i \text{before}_j| \leq \text{now}_i^2 + \text{before}_j^2$, $\sum_{i \in R} e_i \geq 3(\text{den} - \text{num}) \sum_R (\text{now}_i^2 + \text{before}_i^2) - \frac{\text{num}}{2} \sum_{\text{boundary Ports}} (\text{now}_j^2 + \text{before}_j^2)$ over the outside neighbours j : a region's share is negative at most by half a level-square per boundary Port in the current's units.

Derivation S.7 (the tension of a plane wave, Section 4.3). Inputs: the stress reading $G_{aa}(i) = \text{now}_i(\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a}(\text{now}_{i+a} - \text{now}_{i-a})$. Steps: for $\text{now}_i = b \cos(kx_i + \phi)$ along the axis, with $c_n = \cos(kx_i + nk + \phi)$, $G = b^2[c_0 c_2 - c_0^2 - c_1^2 + c_1 c_{-1}]$. The identities $c_0 c_2 = \frac{1}{2}[\cos(2kx + 2k + 2\phi) + \cos 2k]$, $c_1 c_{-1} = \frac{1}{2}[\cos(2kx + 2\phi) + \cos 2k]$, $c_0^2 = \frac{1}{2}[1 + \cos(2kx + 2\phi)]$ and $c_1^2 = \frac{1}{2}[1 + \cos(2kx + 2k + 2\phi)]$ cancel the phase-dependent parts, leaving $G = b^2(\cos 2k - 1) = -2b^2 \sin^2 k$ at every Node and interval. So $T_{aa} = -\text{num} (G(i) + G(i - a)) \text{div } 2 = 2 \text{ num } b^2 \sin^2 k$, and for a wave with the components k_a, k_b the cross reading is $G_{ab} = -2b^2 \sin k_a \sin k_b$.

Measurement (Section 5)

Derivation S.8 (the region's reading factor and the draw's scatter, Section 5.3). Steps: a region of n consecutive rows sums a fringe $\cos(2\pi y/\Lambda)$ to $\text{Re} \sum_{y=0}^{n-1} e^{2\pi i y/\Lambda} = \text{Re} \frac{1 - e^{2\pi i n/\Lambda}}{1 - e^{2\pi i/\Lambda}}$, whose modulus is $\sin(\pi n/\Lambda) / \sin(\pi/\Lambda)$; over n rows the fringe's modulation is kept with the factor $\sin(\pi n/\Lambda) / (n \sin(\pi/\Lambda))$, 0.90 at $n = 4$, $\Lambda = 16$, and 0 at $\Lambda = n$. A frequency read as a click rate over N intervals resolves $\Delta\omega \geq 2\pi/N$, the window's own resolution. N quanta drawn independently by the shares p_i give each region a binomial count, $Np_i \pm \sqrt{Np_i(1 - p_i)}$.

Derivation S.9 (the Wronskian's conservation and its current, Section 5.5). Inputs: the two parts of a plane record step by the same symmetric read matrix $\mathbf{M}$ (the paper's Theorem 3), $\mathbf{re}_{\text{next}} = \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}}$ and the same for $\mathbf{im}$, with the weights $\mathbf{D}$ of the conserved form, $\mathbf{D} \mathbf{M}$ symmetric. Steps: let $W = \mathbf{re}_{\text{now}}^\top \mathbf{D} \mathbf{im}_{\text{before}} - \mathbf{im}_{\text{now}}^\top \mathbf{D} \mathbf{re}_{\text{before}}$. After the step it is $\mathbf{re}_{\text{next}}^\top \mathbf{D} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{next}}^\top \mathbf{D} \mathbf{re}_{\text{now}} = (\mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}})^\top \mathbf{D} \mathbf{im}_{\text{now}} - (\mathbf{M} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{before}})^\top \mathbf{D} \mathbf{re}_{\text{now}}$; the $\mathbf{M}$ terms are $\mathbf{re}_{\text{now}}^\top \mathbf{M}^\top \mathbf{D} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{now}}^\top \mathbf{M}^\top \mathbf{D} \mathbf{re}_{\text{now}}$, which vanish since $\mathbf{M}^\top \mathbf{D} = \mathbf{D} \mathbf{M}$ is symmetric, and the rest is W again. Per Node the same cancellation leaves the Link terms: the weighted Wronskian W_i/p_i^2 changes by $(2 \text{ num}/w) (q_{ij}^2/\Gamma^2) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$ summed over the Node's six Ports at the levels now, the Wronskian's current, which for the integer W_i at the vacuum's paces is $(\text{num}/3 \text{ den}) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$ per Port. Check: a periodic chain of three Nodes at $\Gamma = 2$, $[1, 1]$, $\mathbf{re}_{\text{now}} = (0, 3, 0)$, $\mathbf{im}_{\text{now}} = (3, 0, 0)$ and zeros before steps to $(1, 4, 1)$ and $(4, 1, 1)$ with no remainder; W_0 goes from 0 to 3, W_0/p_0^2 by $3/4$, and the current is $(2/24) \cdot 9 = 3/4$ (an independent automated check's witness); exact

fractions on a chain of five Nodes with contents and Link factors confirm the identity at every Node. For the circular record $\text{re} = A \cos \omega t$, $\text{im} = e A \sin \omega t$ at one Node, $W = A \cos \omega t \cdot e A \sin \omega(t-1) - e A \sin \omega t \cdot A \cos \omega(t-1) = -e A^2 \sin \omega$, constant. Domain: the identity is the line's, $\mathbf{re}_{\text{next}} = \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}}$ exactly. The integer step's division adds $\epsilon = (r - r')/w$ to each part, and the remainder term $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$ is not zero in general: at the vacuum's pace $S = 0$ and $w a_{\text{next}} = 12 \text{num} \Gamma^2 a_{\text{now}} - w a_{\text{before}}$ for a uniform level, so at $[2, 3]$ the parts $(1, 3)$ with zeros before step to $([4/3], [4]) = (1, 4)$ and W goes from 0 to $1 \cdot 3 - 4 \cdot 1 = -1$. An exactly conserved integer charge is open.

Derivation S.10 (Bell's correlation under the declared credit, Section 5.7 and Section 9.4). Inputs: two parts with the products M_1, M_2 of their amplitudes at the two sides' ports; the port coefficients $e(+) = (\cos a, \sin a)$, $e(-) = (-\sin a, \cos a)$ on side A and the same with b on side B; the joint share $J(p, q) = (\sum_k e_k(p) e_k(q) M_k)^2$; $E = \sum_{p,q} s(p)s(q) J(p, q) / \sum_{p,q} J(p, q)$ with $s(\pm) = \pm 1$. Steps: write $c_a = \cos a$, $s_a = \sin a$ and the same for b . The four sums are $M_1 c_a c_b + M_2 s_a s_b$, $M_1 s_a s_b + M_2 c_a c_b$, $-M_1 c_a s_b + M_2 s_a c_b$ and $-M_1 s_a c_b + M_2 c_a s_b$ for $(+, +)$, $(-, -)$, $(+, -)$ and $(-, +)$. Their squares sum to $(M_1^2 + M_2^2)(c_a^2 c_b^2 + s_a^2 s_b^2 + c_a^2 s_b^2 + s_a^2 c_b^2) = M_1^2 + M_2^2$, the cross terms cancelling in pairs. The signed sum is $(M_1^2 + M_2^2)(c_a^2 - s_a^2)(c_b^2 - s_b^2) + 8 M_1 M_2 c_a s_a c_b s_b$, so

$$E(a, b) = \cos 2a \cos 2b + \rho \sin 2a \sin 2b, \quad \rho = \frac{2 M_1 M_2}{M_1^2 + M_2^2},$$

and the marginal $P(A+) = [(M_1 c_a c_b + M_2 s_a s_b)^2 + (-M_1 c_a s_b + M_2 s_a c_b)^2] / (M_1^2 + M_2^2) = (M_1^2 c_a^2 + M_2^2 s_a^2) / (M_1^2 + M_2^2)$, independent of b . For equal parts $\rho = 1$ and $E = \cos 2(a - b)$. The integer settings: with $e(+) = (p, q)$ and $e(-) = (-q, p)$ unnormalised and equal parts, $J(+, +) = J(-, -) = (p_{APB} + q_{AQB})^2$ and $J(+, -) = J(-, +) = (p_{AQB} - q_{APB})^2$, whose sum is $2(p_A^2 + q_A^2)(p_B^2 + q_B^2)$ by Lagrange's identity, so $E = [(p_{APB} + q_{AQB})^2 - (p_{AQB} - q_{APB})^2] / [(p_A^2 + q_A^2)(p_B^2 + q_B^2)]$, a rational number. At $a = (1, 0)$, $a' = (1, 1)$, $b = (12, 5)$, $b' = (5, 12)$: $E(a, b) = (144 - 25)/169 = 119/169$, $E(a, b') = (25 - 144)/169 = -119/169$, $E(a', b) = (17^2 - 7^2)/338 = 120/169$, $E(a', b') = 120/169$, and $S = E(a, b) - E(a, b') + E(a', b) + E(a', b') = 478/169 = 2.8284$. At the ideal angles $S = \sqrt{2}(1 + \rho)$; at the integer settings $S = (238 + 240\rho)/169$. The two local credits: with the pair's phase θ uniform, a credit of each side by the shares of its ports gives $E = \langle \cos 2(a - \theta) \cos 2(b - \theta) \rangle_\theta = \frac{1}{2} \cos 2(a - b)$ and $S = \sqrt{2}$; a credit by the sign of the larger port gives $E = \langle \text{sgn} \cos 2(a - \theta) \text{sgn} \cos 2(b - \theta) \rangle_\theta = 1 - 4|a - b|/\pi$, the sawtooth, whose sum at the four settings is exactly 2. Check: the averages over θ on a grid of 2×10^5 points give $S = 1.4142$ and 2.0000 , and the contrast credit, which discards the pairs near the contrast's zero, $S = 2.8284$ at the efficiency $2/\pi$.

The families (Section 6)

Derivation S.11 (the reach of a field with a gap, Section 6.4). Inputs: a field's rest outside its source, the line with $a_{\text{next}} = a_{\text{before}} = a_{\text{now}} = a$: $6 \text{den } a = \text{num } S_6(a)$ at the vacuum's pace, that is $\text{num}(S_6 - 6)a = 6(\text{den} - \text{num})a$. Steps: for a rest falling along one axis as $e^{-\kappa r}$ and uniform across, $S_6(a) = (2 \cosh \kappa + 4)a$, so $6 \text{den} = \text{num}(2 \cosh \kappa + 4)$ and $\cosh \kappa = 3 \text{den}/\text{num} - 2$. For num just under den , $\cosh \kappa = 1 + \kappa^2/2 + \dots$, so $\kappa^2 = 6 \text{den}/\text{num} - 6 = 6(\text{den} - \text{num})/\text{num}$ and the range is $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$. Links: 20.00 at $[2400, 2401]$; $[1, 2]$ has $\cosh \kappa = 4$, $e^{-\kappa} = 0.127$. The gap is also the rest rotation of the field's own travelling events, $\cos \omega_g = \text{num}/\text{den}$, $\omega_g^2 = 2(\text{den} - \text{num})/\text{den} + \dots$, whose Compton length at light's speed, $c/\omega_g = \sqrt{\text{den}/(6(\text{den} - \text{num}))} = R\sqrt{\text{den}/\text{num}}$, is the range to the gap's first order: Yukawa's identity with no input.

Derivation S.12 (the two speeds, Section 6.5). Steps: light at a content x travels at $(1 - 2x)/\sqrt{3}$ (Derivation S.2). Under the law's line the massless field reads the content as every family does, its own level among it, so its travelling events have the same paces and the two arrive together at every content, the difference of their Shapiro delays 0. With the massless field reading no content, as an earlier version of the implementation had it, its events travel at $1/\sqrt{3}$ and over L Links the two arrive $L\sqrt{3}[1/(1 - 2x) - 1] = 2xL\sqrt{3} + O(x^2)$ intervals apart at long wavelength, against GW170817's bound on $\gamma_{\text{GW}} - \gamma_{\text{EM}}$, between -2.6×10^{-7} and 1.2×10^{-6} ; under the line the measurement bounds no vacuum content.

Bodies (Section 7)

Derivation S.13 (the bound body: the fixed point and the functional, Section 7.2). Inputs: Rule3's line at the paces of the wells the record sources; a standing record $a_{i,t} = \phi_i \cos(\omega_b t)$ with one rotation at every Node; the clock's pace $(p_0/\Gamma)^2 = 1 - 2\Phi_i/\Gamma + \dots$ at a well Φ_i (in levels). The fixed point's equation: with the remainders dropped the line gives at every Node

$$2 \cos \omega_b \phi_i = \frac{1}{w} \left[\sum_a R_a(i) (\phi_{i+a} + \phi_{i-a}) + S_i \phi_i \right] = (\mathbf{M}\phi)_i,$$

so ϕ is an eigenvector of the read matrix $\mathbf{M}$ and the standing record is its top mode, the maximiser of $\phi^\top \mathbf{D} \mathbf{M} \phi$ on the weighted sphere $\phi^\top \mathbf{D} \phi = 1$, $\mathbf{D} \mathbf{M}$ symmetric under the law's read coefficient (S.5) (the generator of Section 7.1 iterates this read act). To the first order in Φ/Γ , $S_i/w = 2 - 2g(1 - 2\Phi_i/\Gamma) - \frac{2\text{num}}{3\text{den}} \sum_a (p_a/\Gamma)^2 = 2 \cos \omega_0 + 4g\Phi_i/\Gamma - \frac{2\text{num}}{3\text{den}} \sum_a (p_a/\Gamma)^2$ and $R_a/w = \frac{\text{num}}{3\text{den}} (p_a/\Gamma)^2$, so, keeping the band's term at the vacuum's Link pace (its product with the well is of the second order),

$$(\mathbf{M}\phi)_i = 2 \cos \omega_0 \phi_i + \frac{4g}{\Gamma} \Phi_i \phi_i - \frac{\text{num}}{3\text{den}} (-\Delta^2 \phi)_i, \quad (-\Delta^2 \phi)_i = 6\phi_i - \sum_a (\phi_{i+a} + \phi_{i-a}).$$

The functional: the well is sourced by the record itself, $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$ for a body of M quanta with the counts $c_j = M\phi_j^2$. Let

$$F[\phi] = \frac{\text{num}}{3\text{den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2,$$

the paper's Eq. (12). Its gradient is $\partial F/\partial \phi_i = \frac{2\text{num}}{3\text{den}} (-\Delta^2 \phi)_i - \frac{8g}{\Gamma} \Phi_i \phi_i$, using the symmetry of G_r (each well counted once, Hartree's form, is what makes the factor $2g$ in F give $4g$ in the gradient). So $\partial F/\partial \phi = 2\lambda\phi$ on the sphere is exactly the fixed point's equation with $\lambda = 2 \cos \omega_0 - 2 \cos \omega_b$: a body is a critical point of F on the sphere, and the virial and the second variation along the dilation are Theorem 4 of the paper. The scaling powers: under $\phi_R(\mathbf{x}) = R^{-d/2} \phi(\mathbf{x}/R)$ the sum over Links of $(\phi_i - \phi_j)^2$ is the lattice form of $\int |\nabla \phi|^2$ and scales as R^{-2} , the pressure K/R^2 . The well energy $\sum_{ij} \phi_i^2 G_{ij} \phi_j^2$ is $\iint \rho(\mathbf{x}) G(\mathbf{x} - \mathbf{y}) \rho(\mathbf{y})$ with $\rho = \phi^2$ of unit mass, which scales as $R^{-(d-2)}$ for the massless kernel $G \sim 1/r$ and for a screened kernel within its reach, and as R^{-d} for a screened kernel beyond its reach, where G acts as a point kernel and the energy is $\int \rho^2$: the powers s_r of Theorem 4. The cloud's size: for the massless kernel in three dimensions $F(R) = a/R^2 - c/R$ with $a = \frac{\text{num}}{3\text{den}} \int |\nabla \phi|^2$ and $c = \frac{2g}{\Gamma} \frac{M}{E} \iint \frac{\rho \rho'}{4\pi |\mathbf{x} - \mathbf{y}|} \cdot 3$ at the unit size, so the minimum is at $R = 2a/c \propto \Gamma E/(gM)$: the radius falls as $1/M$ and grows with Γ , the paper's $2\Gamma/C$ up to the shape's constants, which the generator computes.

Derivation S.14 (the invariant and the count's drift, Eq. (13)). Inputs: the one-Node line $a_{t+1} + a_{t-1} = 2 \cos \omega_t a_t$ with ω_t changing slowly. Steps: at fixed ω the form $D_t = a_t^2 - a_{t+1} a_{t-1} = a_t^2 + a_{t-1}^2 - 2 \cos \omega a_t a_{t-1}$ is conserved and equals $A^2 \sin^2 \omega$ for $a_t = A \cos(\omega t + \phi)$; in the plane (a_t, a_{t-1}) the step is a linear map of determinant 1 whose orbits are the ellipses $D = \text{const}$, of area $\pi D / \sin \omega$. A slow change of ω keeps the area enclosed by the orbit, the action of the map, so $D / \sin \omega = A^2 \sin \omega$ is the adiabatic invariant: the discrete form of E/ω . Equivalently, the ansatz $a_t = C \cos(\sum_{s \leq t} \omega_s + \phi) / \sqrt{\sin \omega_t}$ satisfies the line to the first order in $\omega_{t+1} - \omega_t$. Check: the recurrence stepped 20,000 intervals with ω lowered from 0.841 to 0.600 keeps $D / \sin \omega$ within 4×10^{-5} of its value while D falls by a quarter. The drift: the share at a Node in quanta is $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b / T$ (Eq. (9) at the Node's paces), so with $A^2 \sin \omega_b$ fixed, $N(t_2)/N(t_1) = [\sin \omega_b(t_2) / \sin \omega_b(t_1)] [p_i^2(t_1)/p_i^2(t_2)]$, Eq. (13). Planck's relation: the share $A^2 \sin^2 \omega / T$ is the invariant $I = A^2 \sin \omega / T$ times $\sin \omega$. The body's click: under the declaration that one click is one quantum's change in the absorbing body (Section 7.4), the absorber answers at its own rotation by the two-mode line of S.16, one transition per click from the mode ω_i to the mode ω_j , and the share it takes is $T(\sin \omega_j - \sin \omega_i) = T \sin(\omega_j - \omega_i) [1 - \omega_i \omega_j / 2 + O(\omega^4)]$, by $\sin \omega_j - \sin \omega_i = 2 \cos \frac{\omega_i + \omega_j}{2} \sin \frac{\omega_j - \omega_i}{2}$: one photon's share $T \sin \omega_L$ at the line $\omega_L = \omega_j - \omega_i$, to a relative $\omega_0^2/2 < 10^{-18}$ at nature's gap (S.27). So a body counts photons, I per click, and reads $\hbar \omega$ per click with T for $\hbar$; the share is the energy a click carries, and through a deepening well the number of quanta, I , is kept.

Derivation S.15 (the clock's click rate, Section 7.4). Inputs: Eq. (4) at $k = 0$ with $p_0^2 = \Gamma^2(1 - 2U + 2U^2)$: $\cos \omega(U) = 1 - g(1 - 2U + 2U^2) = \cos \omega_0 + 2gU - 2gU^2$. Steps: to first order $d(\cos \omega) = 2g dU = -\sin \omega_0 d\omega$, so $d\omega/\omega_0 = -[2g/(\omega_0 \sin \omega_0)] dU = -f(\omega_0) dU$ with $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$; $f = 1.063$ at $[2, 3]$, and as $\omega_0 \rightarrow 0$, $1 - \cos \omega_0 \rightarrow \omega_0^2/2$ and $f \rightarrow 1$, the rate $1 - U$ of the clock's pace alone. The light clock: a packet between two declared faces L Links apart at the content x has the round trip $2L/v_g = 2L\sqrt{3}/N^2$ (Derivation S.2), so its rate falls by $2x$ to the first order; between two bodies, which stand L/h Links apart under the principle, the round trip is $2L\sqrt{3}/N$ (Derivation S.31).

Derivation S.16 (the resonance at twice the rotation, Section 7.4). Inputs: the one-Node line with the rotation modulated by a level $b_L \cos \omega_L t$ read into the content, the law's two-mode line; light's declared read is the holder's rotation (Section 5.5), for which this derivation is not made: $2 \cos \omega(t) = 2 - 2g(1 - 2x + 2x^2)$ with $x = x_0 + b_L \cos(\omega_L t)/\Gamma$, so $d(2 \cos \omega)/dx = 4g(1 - 2x)$ and $a_{\text{next}} + a_{\text{before}} = [2 \cos \omega_b + \epsilon \cos(\omega_L t + \vartheta)] a_{\text{now}}$ with $\epsilon = 4g(1 - 2x_0) b_L/\Gamma$. Steps: write $a_t = \text{Re}[z_t e^{i\omega_b t}]$ with z slowly varying, $z_{t \pm 1} = z_t \pm z'$. Then $a_{t+1} + a_{t-1} = \text{Re}[e^{i\omega_b t} (2 \cos \omega_b z_t + 2i \sin \omega_b z')]$, and the product $\epsilon \cos(\omega_L t + \vartheta) \text{Re}[z_t e^{i\omega_b t}]$ has, at $\omega_L = 2\omega_b$, the resonant part $(\epsilon/2) \text{Re}[\bar{z}_t e^{i\vartheta} e^{i\omega_b t}]$. Equating the slowly varying coefficients of $e^{i\omega_b t}$, $2i \sin \omega_b z' = (\epsilon/2) e^{i\vartheta} \bar{z}$, so $z' = -i\epsilon e^{i\vartheta} \bar{z}/(4 \sin \omega_b)$; this antilinear map $z \mapsto \lambda \bar{z}$ with $|\lambda| = \mu$ has the real eigenvalues $\pm \mu$ on two lines of the z plane, so the amplitude grows as $e^{\mu t}$ along one line and the share, quadratic

in the amplitude, as $e^{2\mu t}$. Detuned by $\Delta = \omega_L - 2\omega_b$, the substitution $z = \zeta e^{i\Delta t/2}$ gives $\zeta' = -i(\Delta/2)\zeta - i\mu e^{i\theta}\bar{\zeta}$, whose exponents are $\pm\sqrt{\mu^2 - \Delta^2/4}$: growth only for $|\Delta| < 2\mu$, a full width $\epsilon/\sin\omega_b$. Check: the recurrence stepped 20,000 intervals at $2\cos\omega_b = 1.4317$ and $\epsilon = 0.013$ grows at 0.00461 per interval against the formula's first-order 0.004654, the difference of the order of ϵ . The two-mode line: with two bound modes the same modulation multiplies $a_t = \text{Re}[z_i e^{i\omega_i t} + z_j e^{i\omega_j t}]$, and its product with $\cos\omega_L t$ carries the frequencies $\omega_i \pm \omega_L$. The term resonant with the mode j from the mode i is at $\omega_L = \omega_j - \omega_i$, a transfer between the two modes, and the term that drives both from the conjugate of the other is at $\omega_L = \omega_i + \omega_j$, of which the one-mode case above is the degenerate case $i = j$ (at $i = j$ the difference line is $\omega_L = 0$, a static shift and no line). The transfer's rate, $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin\omega_i \sin\omega_j})$, follows by the same slowly varying coefficients with the two modes' $2i \sin\omega_i z'_i$ and $2i \sin\omega_j z'_j$ coupled through ϵ_{ij} , a Rabi oscillation; under light's declared read, the rotation, the derivation is open.

Derivation S.17 (the cross current of two standing records, Section 7.5). Inputs: two standing records $a_t = A \cos(\omega_1 t + \alpha)$ and $b_t = B \cos(\omega_2 t + \beta)$ at the two ends of a Link. Steps: the current $\text{num}(a_t b_{t-1} - a_{t-1} b_t)$ is $\text{num}AB[\cos(\omega_1 t + \alpha)\cos(\omega_2 t - \omega_2 + \beta) - \cos(\omega_1 t - \omega_1 + \alpha)\cos(\omega_2 t + \beta)]$, a sum of cosines of $(\omega_1 - \omega_2)t$ and $(\omega_1 + \omega_2)t$ with the constant parts $\frac{1}{2}\cos(\alpha - \beta + \omega_2)$ and $\frac{1}{2}\cos(\alpha - \beta - \omega_1)$ only when $\omega_1 = \omega_2$; for $\omega_1 \neq \omega_2$ its mean over the beat is 0, so no quantum passes at the first order. The reach: two bodies' tails are $a_1 e^{-\kappa d_1}$ and $a_2 e^{-\kappa d_2}$ at a Link $d = d_1 + d_2$ Links from the two centres, so the current there is of the order $a_1 a_2 e^{-\kappa d}$ per interval and falls under one quantum, T , at $d = \ln(a_1 a_2 / T) / \kappa$.

The forms of nature (Section 8)

Derivation S.18 (the fall and the Newtonian potential the orbits measure, Eq. (14)). Inputs: Eq. (4) at a content U for a reader at the weight 1, $\cos\omega(k, U) = 1 - g(1 - 2U + 2U^2) - \frac{\text{num}}{3\text{den}}(1 - 2U)^2(1 - \cos k)$ along the motion. Steps: at small k and U , $\omega(k, U) = \omega_0(U) + k^2/(2m^*)$ with the inertia $m^* = 3\text{den} \sin\omega_0 / \text{num} = 3 \tan\omega_0$ (from $\sin\omega_0 d\omega = \frac{\text{num}}{3\text{den}} k dk$), so the group velocity is $v = k/m^*$. The rays of $\omega(k, x)$ give $dk/dt = -\partial\omega/\partial x = -(d\omega_0/dU) \nabla U = (2g/\sin\omega_0) |\nabla U|$ toward the well (Derivation S.15), so the acceleration is

$$a = \frac{1}{m^*} \frac{dk}{dt} = \frac{\text{num}}{3\text{den} \sin\omega_0} \cdot \frac{2g}{\sin\omega_0} |\nabla U| = \frac{2 \cos\omega_0 (1 - \cos\omega_0)}{3 \sin^2\omega_0} |\nabla U| = \frac{2 \cos\omega_0}{3(1 + \cos\omega_0)} |\nabla U|,$$

which in the gap $\epsilon = 1 - \cos\omega_0$ is $[(1 - \epsilon)/(3(1 - \epsilon/2))] |\nabla U|$: 0.2667 at $[2, 3]$ and $1/3$ as the gap closes. Nature's fall is $a = c^2 |\nabla U_K|$ with $U_K = GM_K/(c^2 r)$ and $c^2 = 1/3$ here, so $U_K = [2 \cos\omega_0/(1 + \cos\omega_0)] U$, Eq. (14), 0.800 at $[2, 3]$.

Derivation S.19 (the post-Newtonian parameters against the Newtonian potential the orbits measure, Section 8). Inputs: Eq. (14), $U = \lambda U_K$ with $\lambda = (1 + \cos\omega_0)/(2 \cos\omega_0) = (\text{den} + \text{num})/(2\text{num})$; the three forms in U : the bending $4U_b$, the periapsis $6\pi U_{1,1}/(a(1 - e^2))$, the redshift $f(\omega_0) U$. Steps: the bending is $4\lambda U_{K,b} = 2(1 + \text{den}/\text{num}) U_{K,b}$, and the post-Newtonian bending is $(1 + \gamma) 2GM/(c^2 b) = (1 + \gamma) 2U_{K,b}$, so $1 + \gamma_K = 1 + \text{den}/\text{num}$, $\gamma_K = \text{den}/\text{num}$. The periapsis is λ times Einstein's $6\pi U_{K,1}/(a(1 - e^2))$, and the post-Newtonian advance is $(2 - \beta + 2\gamma)/3$ of Einstein's, so $(2 - \beta_K + 2 \text{den}/\text{num})/3 = (\text{den} + \text{num})/(2\text{num})$, whence $\beta_K = 2 + 2 \text{den}/\text{num} - 3(\text{den} + \text{num})/(2\text{num}) = (\text{den} + \text{num})/(2\text{num})$. The redshift is $f(\omega_0) \lambda U_K$, and $f\lambda = \frac{2(1 - \cos\omega_0)}{\omega_0 \sin\omega_0} \cdot \frac{1 + \cos\omega_0}{2 \cos\omega_0} = \frac{\sin^2\omega_0}{\omega_0 \sin\omega_0 \cos\omega_0} = \frac{\tan\omega_0}{\omega_0}$, so $1 + \alpha_K = \tan\omega_0/\omega_0$. At $[2, 3]$: 1.5, 1.25 and 1.329; all three tend to 1 as $\omega_0 \rightarrow 0$, and $\gamma_K - 1 < 2 \times 10^{-5}$ requires $\text{den}/\text{num} - 1 < 2 \times 10^{-5}$.

Derivation S.20 (Shapiro's delay, Section 8 (c)). Inputs: light's index $n = 1 + 2U$ with $U = \mu/r$ (Derivation S.2). Steps: the delay along a straight path at the impact parameter b from $-x_1$ to x_2 is $\int (n - 1) dl/c = 2\sqrt{3} \mu \int_{-x_1}^{x_2} dl/\sqrt{l^2 + b^2} = 2\sqrt{3} \mu [\text{arsinh}(x_2/b) + \text{arsinh}(x_1/b)]$, which for $x_1, x_2 \gg b$ is $2\sqrt{3} \mu \ln(4x_1 x_2/b^2)$: Shapiro's delay with $\gamma = 1$ in U , and with γ_K against U_K by the same factor as the bending, since the bending is the transverse derivative of the same integral.

Derivation S.21 (the moving clock to the second order, Section 8 (e)). Inputs: the exact band of a free record of the pair $[\text{num}, \text{den}]$ along an axis, from Eq. (1) at the vacuum's pace (S.1): $\cos\omega = c_0(2 + \cos k)$ with $c_0 = \text{num}/(3\text{den})$, $\cos\omega_0 = 3c_0$; the rotation at the moving centre, $\Omega(k) = \omega(k) - k\omega'(k)$, the phase rate less k times the group velocity $v = \omega'(k)$, which is what a clock carried by the packet reads. Steps: write $s = \sin\omega_0$, $c = \cos\omega_0$ and $u = 1 - \cos k = k^2/2 - k^4/24 + \dots$; then $\cos\omega = c - c_0 u$, and with $\omega = \omega_0 + \delta$, $\cos\omega = c - s\delta - c\delta^2/2 + \dots$, so $s\delta + c\delta^2/2 = c_0 u$ and

$$\omega(k) = \omega_0 + \alpha k^2 - \beta k^4 + O(k^6), \quad \alpha = \frac{c_0}{2s}, \quad \beta = \frac{c_0}{24s} + \frac{c c_0^2}{8s^3}.$$

Hence $v = \omega' = 2\alpha k - 4\beta k^3$, $\Omega = \omega - k\omega' = \omega_0 - \alpha k^2 + 3\beta k^4$, and inverting $v(k)$, $k^2 = v^2/(4\alpha^2) + \beta v^4/(4\alpha^5) + \dots$, so

$$\frac{\Omega}{\omega_0} = 1 - \frac{v^2}{4\alpha\omega_0} - \frac{\beta v^4}{16\alpha^4\omega_0} + O(v^6) = 1 - \frac{v^2}{2c_m^2} - \frac{2\omega_0}{\sin 2\omega_0} \cdot \frac{v^4}{8c_m^4} + O(v^6), \quad c_m^2 = 2\alpha\omega_0 = \frac{c_0 \omega_0}{s} = \frac{\omega_0}{3 \tan\omega_0},$$

where the fourth-order coefficient is $\beta/(16\alpha^4\omega_0)$ over Lorentz's $1/(8c_m^4)$, which reduces with $3c_0 = c$ to $\omega_0(s/c + c/s) = \omega_0/(\sin\omega_0 \cos\omega_0) = 2\omega_0/\sin 2\omega_0$. So the tick of a free quantum is Lorentz's factor at the family's own light speed c_m to the second order in v exactly, and at the fourth order it carries the factor $2\omega_0/\sin 2\omega_0$, 1.69 at [2, 3] and 1 as the gap closes: the full factor is reached only in the limit of a small rest rotation, the same limit in which the post-Newtonian parameters reach 1 (S.19). Check: with the exact band at [2, 3] the residual $(f - 1 + v^2/(2c_m^2))/v^4$ at $v = 0.02$ is -3.37 against 1.6926 times Lorentz's -1.988 , that is -3.365 . A body: a bound body's centre moves on the body's own band, $\omega_b(k)$, whose rest rotation and curvature the generator computes from the fixed point (S.13); the same steps on that band give the same form with the body's $c_b^2 = \omega_b/m_b^*$, and the body's band is a computed input, not derived in closed form here.

Derivation S.22 (de Broglie's relation, Section 8 (h)). Steps: from Derivation S.18, $v = k/m^*$ with $m^* = 3 \tan \omega_0$ at the vacuum's content, so $k = m^*v = \omega_b v/c_m^2$ with $c_m^2 = \omega_0/m^* = \omega_0/(3 \tan \omega_0)$, the family's own light speed; 0.2508 at [2, 3].

The two slits (Section 9)

Derivation S.23 (the fringe's spacing and the near-field minima, Section 9.2). Inputs: the gaps centred at the rows 18 and 30, $d = 12$; the screen $L = 24$ Links from the wall; $\lambda = 8$. Steps: the far-field spacing is $\lambda L/d = 16$ Links. In the near field the minima are where the two gaps' paths to the screen differ by half a wavelength, $\sqrt{L^2 + (y - 18)^2} - \sqrt{L^2 + (y - 30)^2} = \pm 4$, whose solutions are $y = 15.28$ and $y = 32.72$: the rows 15 and 33. Check: the two roots computed from the equation above.

Calibration (Section 10)

Derivation S.24 (the Link's bound from light's dispersion, Section 10.2). Inputs: the group speed $v_g = (1/\sqrt{3})(1 - k^2/12)$ (Derivation S.1); the bound on a quadratic dependence of light's speed on its energy, MAGIC's $E_{QG,2}$ above about 6×10^{10} GeV, that is $|\delta v/v| = (3/2)(E/E_{QG,2})^2 < 4.5 \times 10^{-16}$ at $E = 1$ TeV in the standard quadratic form. Steps: $k^2/12 < 4.5 \times 10^{-16}$ gives $k < 7.3 \times 10^{-8}$ per Link at 1 TeV; with $k = 2\pi \text{Link}/\lambda$ and $\lambda = 1.24 \times 10^{-18}$ m at 1 TeV, the Link is below $k\lambda/(2\pi) = 1.4 \times 10^{-26}$ m.

Derivation S.25 (Newton's constant as a reading, Eq. (18)). Inputs: the massless field's rest far from a static source of s quanta per interval, $3s/(4\pi r)$ over the divisor E_g ; the clock's share $U = c/\Gamma$. Steps: $U = 3s/(4\pi r \Gamma E_g)$, and nature's $U = GM/(c^2 r)$ with $M = s$ in quanta and $c^2 = 1/3$ gives $G/c^2 = 3/(4\pi \Gamma E_g)$, that is $G_{\text{clock}} = 1/(4\pi \Gamma E_g)$; Kepler's G , read from the orbits of matter, is $[2 \cos \omega_0/(1 + \cos \omega_0)] G_{\text{clock}}$ by Eq. (14).

Derivation S.26 (the fine-structure constant as a reading, Section 10.1). Inputs: a circular record of the plane at the vacuum's paces, $z = Ae^{i\omega_0 t}$, whose two parts' shares sum to the count, $2A^2 \sin^2 \omega_0 = T$ for one quantum (Section 4.4, Section 5.5), and whose Wronskian is $|W| = A^2 \sin \omega_0$ (S.9). The holder of the sign's rest far from a static source of s per interval, $3s/(4\pi r)$ at the divisor 1 (Derivation S.25's line at $E_s = 1$). The rotation's angle $\theta = \text{level}/\Gamma$ (Section 5.5). De Broglie's $k = p/\hbar$ (S.22) and light's $c = 1/\sqrt{3}$ (S.1). Steps: one quantum writes $s = W/T = A^2 \sin \omega_0/(2A^2 \sin^2 \omega_0) = 1/(2 \sin \omega_0)$ per interval. The angle at r is $\theta(r) = 3s/(4\pi r \Gamma)$. A reader of unit sense gains the angle per interval, so its wave number changes by the gradient, $dk/dt = -\nabla \theta = 3s/(4\pi r^2 \Gamma)$; Coulomb's law in nature's units, $dk/dt = \alpha c/r^2$, gives $\alpha = r^2(dk/dt)/c = \sqrt{3} r^2 dk/dt = 3\sqrt{3} s/(4\pi \Gamma) = (3\sqrt{3}/8\pi)/(\Gamma \sin \omega_0)$. The level weight LW of the holder enters the source and the read alike and cancels. Check: $3\sqrt{3}/(8\pi) = 0.20675$; $137.036 \times 0.20675 = 28.33$. With the pair $[\text{den} - 1, \text{den}]$, $\sin^2 \omega_0 = 1 - (1 - 1/\text{den})^2 = 2/\text{den} - 1/\text{den}^2$, so $\text{den} = 2/\omega_0^2$ to the first order and $\text{den} = 2\Gamma^2/28.33^2 = \Gamma^2/401.3$.

Derivation S.27 (the units, the highest photon energy and the matter pair's gap, Section 10.1). Inputs: the electron as the free quantum of matter, rotating at ω_0 per interval, and nature's electron at $m_e c^2/\hbar$ per second; light's band $\cos \omega = 1 - \frac{1}{3} \sum_a (1 - \cos k_a)$ (S.1); the highest photon observed, 1.4 PeV. Steps: one interval is $\omega_0 \hbar/(m_e c^2) = \omega_0 \times 1.288 \times 10^{-21}$ s, and one Link, light moving $1/\sqrt{3}$ Link per interval at c , is $\sqrt{3} \omega_0 \bar{\lambda}_e = \omega_0 \times 6.69 \times 10^{-13}$ m with $\bar{\lambda}_e = \hbar/(m_e c) = 3.862 \times 10^{-13}$ m. The band's top along the body diagonal, $k_a = \pi$ for all a , is $\cos \omega = -1$, $\omega = \pi$, and along an axis, $k = \pi$ alone, $\cos \omega = 1/3$, $\omega = 1.231$. Along an axis the group velocity vanishes at the edge; along the diagonal, $k_a = q$ on every axis, the band is $\omega = q$ exactly, so the speed stays $1/\sqrt{3}$ up to the corner, where the band has a cusp and no one gradient. So the highest energy a light record carries is $\pi m_e c^2/\omega_0 = 1.605 \text{ MeV}/\omega_0$ along the diagonal and $1.231 m_e c^2/\omega_0 = 0.629 \text{ MeV}/\omega_0$ along an axis. A photon of 1.4 PeV that arrived, with no orientation of the lattice known, needs the diagonal's ceiling above it, $1.605 \text{ MeV}/\omega_0 > 1.4 \times 10^9 \text{ MeV}$, $\omega_0 < 1.1 \times 10^{-9}$. Then $\text{den} = 2/\omega_0^2 > 1.5 \times 10^{18}$, $\Gamma = 28.33/\omega_0 > 2.5 \times 10^{10}$ (Derivation S.26), and

$\gamma_K - 1 = 1/\cos\omega_0 - 1 \approx \omega_0^2/2 < 6.6 \times 10^{-19}$ (S.19). Were the photon's path along an axis, the axis ceiling gives $\omega_0 < 4.5 \times 10^{-10}$, $\text{den} > 10^{19}$, $\Gamma > 6.3 \times 10^{10}$ and $\gamma_K - 1 < 1.0 \times 10^{-19}$, a sufficient bound and not a necessary one. Check: $\hbar/(m_e c^2) = 1.0546 \times 10^{-34}/8.187 \times 10^{-14} = 1.288 \times 10^{-21}$ s; $\sqrt{3} \times 3.862 \times 10^{-13} = 6.69 \times 10^{-13}$ m; $\pi \times 0.511 = 1.605$; $1.231 \times 0.511 = 0.629$; $1.605/1.4 \times 10^{-9} = 1.15 \times 10^{-9}$; $[\pi - \omega(\pi - \epsilon)]/(\sqrt{3}\epsilon) = 0.5774$ at $\epsilon = 10^{-2}$ and 10^{-3} on the diagonal.

Derivation S.28 (the content's index for every family, Section 8 (p)). Inputs: Eq. (4) at the vacuum's pace and at a content U under the composed paces, $(p_0/\Gamma)^2 = N^2$ and $(p_a/\Gamma)^2 = N^4$ with $N = (1 - 1/\Gamma)^c = e^{-U}$ to $O(1/\Gamma)$ (Derivation S.2); a stationary boundary conserves the rotation ω . Steps: outside, $\cos\omega = 1 - g - \frac{\text{num}}{3\text{den}}(1 - \cos k)$ with $g = 1 - \text{num}/\text{den}$; inside, $\cos\omega = 1 - gN^2 - \frac{\text{num}}{3\text{den}}N^4(1 - \cos k_{\text{in}})$. Equating, $\frac{\text{num}}{3\text{den}}N^4(1 - \cos k_{\text{in}}) = \frac{\text{num}}{3\text{den}}(1 - \cos k) + g(1 - N^2)$, so $1 - \cos k_{\text{in}} = [1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}]/N^4$, exact in k on the domain where the right side lies in $[0, 2]$. With $1 - N^2 = 2U - 2U^2 + \dots$ and $N^{-4} = 1 + 4U + \dots$ the first order is $[1 - \cos k + 6(\text{den} - \text{num})U/\text{num}]/(1 - 2U)^2$, the quadratic forms' line, the mass term's second order $-6(\text{den} - \text{num})U^2/\text{num}$. For light, $g = 0$: $1 - \cos k_{\text{in}} = (1 - \cos k)/N^4$, so $\sin(k_{\text{in}}/2) = \sin(k/2)/s$ with $s = N^2$, exact in the law's variable $2\sin(k/2)$ where $\sin(k/2) \leq s$. The phase index is $k_{\text{in}}/k = (2/k) \arcsin[\sin(k/2)/s] = 1/s + k^2(1/s^3 - 1/s)/24 + O(k^4)$, which is $1/s = e^{2U}$, Derivation S.2's long-wavelength index, at $k \rightarrow 0$ alone. At $k = \pi/2$ and $s = 0.8$ the exact ratio is 1.380 against $1/s = 1.25$, and to first order in U the index is $1 + [4 \tan(k/2)/k]U$, the coefficient 2 at $k \rightarrow 0$ and $8/\pi$ at $k = \pi/2$: the lattice's own chromatic dispersion, absent in the ray limit in which the bending, the delay and the lens are read. For slow matter, $k, k_{\text{in}} \ll 1$: $k_{\text{in}}^2 = [k^2 + 6(\text{den} - \text{num})(1 - N^2)/\text{num}]/N^4$, so $n_m^2 = k_{\text{in}}^2/k^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num}k^2)]/N^4$, to first order $[1 + 12(\text{den} - \text{num})U/(\text{num}k^2)]/(1 - 2U)^2$. Check: with $k^2 = m^*v^2$ and $m^* = 3 \tan\omega_0$ (S.22), the $1/k^2$ term is $12(\text{den} - \text{num})U/(\text{num}m^*v^2) = [4(1 - \cos\omega_0)/(3 \cos\omega_0 \tan^2\omega_0)]U/v^2$, which as $\omega_0 \rightarrow 0$ is $(2/3)U/v^2 = 2(U/3)/v^2 = 2\Phi/v^2$ with $\Phi = c^2U$, $c^2 = 1/3$: nature's index for a slow body in a potential. The factor $6(\text{den} - \text{num})/\text{num}$ over m^*2 is the matter band's own, the Kepler column of Eq. (14), and $1/N^4$ the Link's double read.

Derivation S.29 (the clusters' cosmology: the ball of dust, Section 7.5). Inputs: Newton's pull between bodies at Kepler's G_K (Section 8 (j), Derivation S.25); a finite ball of dust of radius $a(t)$ and uniform density ρ on the static lattice; the Doppler period of a receding emitter, row (m), and the moving clock's factor, row (e). Steps: a shell at a feels the mass inside it, $M(a) = (4\pi/3)\rho a^3$, and nothing from outside (Newton's theorem on a sphere, which holds for the $1/r$ field of Section 8 (j)), so $\ddot{a} = -G_K M(a)/a^2 = -(4\pi G_K/3)\rho a$. Multiplying by $\dot{a}$ and integrating with ρa^3 constant gives $\dot{a}^2/a^2 = (8\pi G_K/3)\rho - \kappa/a^2$ with κ the integration constant, positive where the ball is bound: Friedmann's two equations for dust, as Milne and McCrea derived them for the Newtonian ball. The deceleration parameter is $q_0 = -\ddot{a}a/\dot{a}^2 = (4\pi G_K \rho/3)/H^2 = \Omega_m/2$ with $\Omega_m = 8\pi G_K \rho/(3H^2)$, positive for every $\rho > 0$. The redshift: the period of a receding emitter at the speed βc is read $(1 + \beta)$ times longer at the resting detector (row (m)) and the emitter's own tick is longer by the moving clock's factor $1/\sqrt{1 - \beta^2}$ at $c_m = c$ (row (e)), so $1 + z = (1 + \beta)/\sqrt{1 - \beta^2} = 1 + \beta + \beta^2/2 + \dots$, nature's relativistic Doppler form to the second order, the classical $1/(1 - \beta) = 1 + \beta + \beta^2$ differing at the second order. At the computed pair $c_m = 0.867c$ the second order differs, as every number of the computed pair does. Every click's duration is stretched by the same $1 + z$, the light curves' time dilation, and a surface's brightness falls by $(1 + z)^{-4}$, Tolman's dimming, both from the same two rows; light from a source at r falls into the ball's deeper centre, a blueshift $\Delta U = (\Omega_m/4)(H_0 r/c)^2$ beside the Doppler, the second order that makes $d_L(z)$ Friedmann's with q_0 unchanged. The vacuum's write: a uniform source s in the massless row's rest, $\nabla^2 L = 3s$ in the lattice's units ($c^2 = 1/3$), has the solution $L = L_{\text{faces}} - s(R^2 - r^2)/2$ on the ball, a bowl rising outward. Its pull on a shell at a is $+\Lambda c^2 a/3$ with Λ proportional to the write's rate $|q_{\text{vac}}|/(E_v \Gamma LW)$, so $\ddot{a} = -(4\pi G_K/3)\rho a + \Lambda c^2 a/3$ and $\dot{a}^2/a^2 = (8\pi G_K/3)\rho + \Lambda c^2/3 - \kappa/a^2$, Friedmann's equations with a cosmological constant; $q_0 = \Omega_m/2 - \Omega_\Lambda$, and $\ddot{a} = 0$ where $\rho = 2\rho_\Lambda$, $1 + z_t = (2\Omega_\Lambda/\Omega_m)^{1/3}$, $z_t = 0.67$ at $\Omega_\Lambda = 0.7$. Light's band at (π, π, π) needs $L \geq 0$ at every Node (the guard), so $L_{\text{faces}} \geq sR^2/2$, in U the faces' content $\Lambda R^2/6$; at one astronomical unit the bowl's pull $\Lambda r/3$ is 2×10^{-22} of the Sun's.

Derivation S.30 (the shadow's capture radius from light's index, Section 8). Inputs: light's index $n(r) = e^{2U}$ with $U = m/r$ (Derivation S.2 under the composed paces), $m = GM/c^2$ in Links. Steps: in a static isotropic index a ray at the impact parameter b keeps $n(r)r \sin\psi = b$ along its path, so a circular ray stands where $d(nr)/dr = 0$: $d(re^{2m/r})/dr = e^{2m/r}(1 - 2m/r) = 0$ at $r = 2m$, the photon sphere, and the capture radius is $b = n(2m)2m = 2em = 5.437m$. Schwarzschild's is $3\sqrt{3}m = 5.196m$, so the shadow is larger by $2e/(3\sqrt{3}) - 1 = 0.046$. Check: with the quadratic index $1/(1 - 2U)$ the same condition gives $r = 4m$ and $b = 8m$, +54 percent, which the Event Horizon Telescope excludes; the composed index is the one the principle gives.

Derivation S.31 (the two clocks under the principle, Section 10.2 (4)). Inputs: the principle of Section 3.4, a uniform level U a change of units, time by N and length by h with $hN = 1$; light's speed N^2 (S.2). Steps: a bound body in the level is the flat body in the local units, so its size is $1/h = N$ Links. Its band at the Node is $2 - 2\cos\omega = 2(1 - \nu)N^2 + (\nu/3)N^4 \sum_a s(k_a)^2$ with $\nu = \text{num}/\text{den}$ and $s(x) = 2\sin(x/2)$, since $(p_0/\Gamma)^2 = N^2$ and $(p_a/\Gamma)^2 = N^4$, and its lattice wave numbers are $k = k_{\text{proper}}/N$, so $(\nu/3)N^4 \sum_a s(k_a)^2 = (\nu/3)N^2 \sum_a k_{\text{proper},a}^2 [1 - k_a^2/12 + \dots]$:

every mode of the record, the rest rotation and every level of its spectrum, scales by N in $s(\omega)$, exactly at long wavelength and to the lattice's $k^2/12$ at finite k . In ω the rest line's rate is $-2\tan(\omega_0/2)/\omega_0 = -f(\omega_0)$ per unit U , Derivation S.15's number. A difference $s(\omega'_2) - s(\omega'_1)$ scales by N exactly and is no observable. The temporal beat a detector reads, $\Delta' = \omega'_2 - \omega'_1$ in $\cos(\Delta't)$, with $\omega' = 2 \arcsin[N \sin(\omega/2)] = N\omega [1 - (1 - N^2)\omega^2/24 + O(\omega^4)]$, is $\Delta' = N\Delta [1 - (1 - N^2)(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)/24 + O(\omega^4)]$: at $N = 0.8$, $\omega_1 = 0.4$ and $\omega_2 = 0.9$ the exact beat is 0.3916 against $N\Delta = 0.4$ and the series 0.3920. The relative correction is 2.5×10^{-38} at nature's gap (ω of 10^{-14} , $U = 10^{-9}$) and 1.4 percent at the engine's $\omega_b = 0.78$ and $U = 0.1$, the lattice's arithmetic and not the law's, which the engine's blind reads in both variables. A cavity whose mirrors are bodies stands $L/h = LN$ Links apart and its round trip is $2LN\sqrt{3}/N^2 = 2L\sqrt{3}/N$ intervals, the rate N exactly with no ω correction, the atom's beat to its correction $(1 - N^2)\omega^2/24$ above; a cavity of declared faces keeps its L Links and ticks at N^2 . So under the principle no clock type is singled out, which is what the null redshift tests measure, while a cavity of declared faces is an instrument's Node and no body of nature.

Derivation S.32 (the conserved form's Link factor, two witnesses, Theorem 3). Inputs: *Theorem 3's line without its division on a periodic board at the content 0 with every Link's factor q , so $p_0 = p_i = \Gamma$, $R = 2 \text{ num } q^2$, $w = 6 \text{ den } \Gamma^2$ and $S = 12 \text{ num } \Gamma^2 - 6R$; the form E of Eq. (6) and the same form with the plain Link term $-2 \text{ num } \sum_i \text{now}_i S_6(\text{before})_i$; exact fractions, no run of the engine.* First witness: *three Nodes along x with y and z folded, $\text{num} = \text{den} = 1$, $\Gamma = 4$, $q = 2$ ($R = 8$, $w = 96$, $S = 144$), $\text{now} = (1, 0, 0)$ and $\text{before} = 0$: one step gives $\text{next} = (11/6, 1/12, 1/12)$; the plain form goes from 6 to $-21/4$ and Eq. (6) stays 6.* Second witness: *3^3 Nodes at $[2, 3]$, $\Gamma = 10$, $q = 5$ ($R = 100$, $w = 1,800$, $S = 1,800$), a uniform $\text{now} = 3$ over $\text{before} = 0$: the step gives $\text{next} = 4$ at every Node with the remainder 0, so the integer step and the line agree; the plain form per Node goes from 162 to -54 and Eq. (6) stays 162 (4,374 over the board). Both are an independent automated check's, recomputed here. Check: at $q = \Gamma$ the factor is 1 and the two forms coincide, the domain of the plain current.*